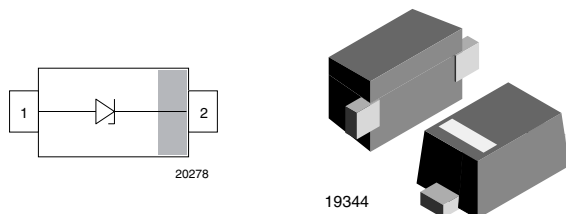


Single ESD-Protection Diode in SOD-523



MARKING (example only)



Bar = cathode marking

X = date code

Y = type code (see table below)

LINKS TO ADDITIONAL RESOURCES



3D Models

FEATURES

- Compact SOD-523 package
- Low package height < 0.7 mm
- 1-line ESD-protection
- AEC-Q101 qualified available
- Working range 0 V to 5 V
- Low leakage current $I_R < 0.1 \mu A$
- Surge current 16 A (at 8/20 μs)
- ESD-protection acc. IEC 61000-4-2
± 30 kV contact discharge
± 30 kV air discharge
- e3 - Sn
Tin plated exposed side wall of lead frame
- Soldering can be checked by standard vision inspection
- AOI = automated optical inspection
- No X-ray necessary
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912



ORDERING INFORMATION

PART NUMBER (EXAMPLE)	AEC-Q101 QUALIFIED	ENVIRONMENTAL AND QUALITY CODE			ORDERING CODE (EXAMPLE)
		RoHS COMPLIANT + LEAD (Pb)-FREE TERMINATIONS	TIN PLATED	8K PER 7" REEL (8 mm TAPE)	
				MOQ = 8K/BOX	
VESD05A1-02V	-	G	3	-08	VESD05A1-02V-G3-08
VESD05A1-02V	H	G	3	-08	VESD05A1-02VHG3-08

PACKAGE DATA

DEVICE NAME	PACKAGE NAME	TYPE CODE	WEIGHT	MOLDING COMPOUND FLAMMABILITY RATING	MOISTURE SENSITIVITY LEVEL	SOLDERING CONDITIONS
VESD05A1-02V (-G3-08 / HG3-08)	SOD-523	. J	1.32 mg	UL 94 V-0	MSL level 1 (according J-STD-020)	Peak temperature max. 260 °C

ABSOLUTE MAXIMUM RATINGS

PARAMETER	TEST CONDITIONS	SYMBOL	VALUE	UNIT
Peak pulse current	Acc. IEC 61000-4-5, 8/20 μs /single shot	I_{PPM}	16	A
Peak pulse power	Acc. IEC 61000-4-5, 8/20 μs /single shot	P_{PP}	192	W
ESD immunity	Contact discharge acc. IEC 61000-4-2; 10 pulses	V_{ESD}	± 30	kV
	Air discharge acc. IEC 61000-4-2; 10 pulses		± 30	kV
Operating temperature	Junction temperature	T_J	-55 to +150	°C
Storage temperature		T_{stg}	-55 to +150	°C

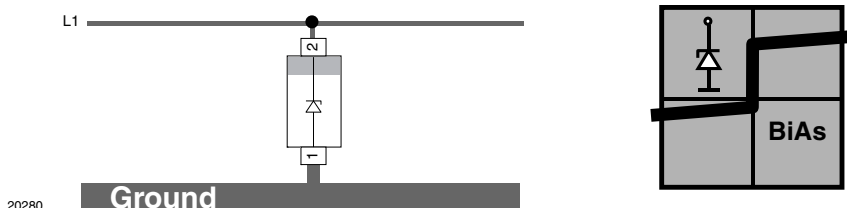
BiAs-MODE (bidirectional asymmetrical protection mode)

With the VESD05A1-02V one signal- or data-lines (L1) can be protected against voltage transients. With pin 1 connected to ground and pin 2 connected to a signal- or data-line which has to be protected. As long as the voltage level on the data- or signal-line is between 0 V (ground level) and the specified maximum reverse working voltage (V_{RWM}) the protection diode between data line and ground offers a high isolation to the ground line. The protection device behaves like an open switch.

As soon as any positive transient voltage signal exceeds the break through voltage level of the protection diode, the diode becomes conductive and shorts the transient current to ground. Now the protection device behaves like a closed switch. The clamping voltage (V_C) is defined by the breakthrough voltage (V_{BR}) level plus the voltage drop at the series impedance (resistance and inductance) of the protection device.

Any negative transient signal will be clamped accordingly. The negative transient current is flowing in the forward direction of the protection diode. The low forward voltage (V_F) clamps the negative transient close to the ground level.

Due to the different clamping levels in forward and reverse direction the VESD05A1-02V clamping behaviour is bidirectional and asymmetrical (BiAs).


ELECTRICAL CHARACTERISTICS ($T_{amb} = 25\text{ }^{\circ}\text{C}$, unless otherwise specified)

PARAMETER	TEST CONDITIONS/REMARKS	SYMBOL	MIN.	TYP.	MAX.	UNIT
Protection paths	Number of lines which can be protected	$N_{channel}$	-	-	1	lines
Reverse stand off voltage	Max. reverse working voltage	V_{RWM}	-	-	5	V
Reverse voltage	At $I_R = 1\text{ }\mu\text{A}$	V_R	5	-	-	V
Reverse current	At $V_R = 5\text{ V}$	I_R	-	< 0.1	1	μA
Reverse breakdown voltage	At $I_R = 1\text{ mA}$	V_{BR}	6	6.8	7.5	V
Reverse clamping voltage	At $I_{PP} = 1\text{ A}$; 8/20 μs test pulse	V_C	-	7.2	8.5	V
	At $I_{PP} = I_{PPM} = 16\text{ A}$; 8/20 μs test pulse	V_C	-	10.5	12	V
Forward clamping voltage	At $I_{PP} = 0.2\text{ A}$; 8/20 μs test pulse	V_F	-	0.88	1.1	V
	At $I_{PP} = 1\text{ A}$; 8/20 μs test pulse	V_F	-	1	1.5	V
	At $I_{PP} = I_{PPM} = 16\text{ A}$; 8/20 μs test pulse	V_F	-	3.2	4.5	V
Capacitance	At $V_R = 0\text{ V}$; $f = 1\text{ MHz}$	C_D	-	130	150	pF
	At $V_R = 2.5\text{ V}$; $f = 1\text{ MHz}$	C_D	-	76	-	pF

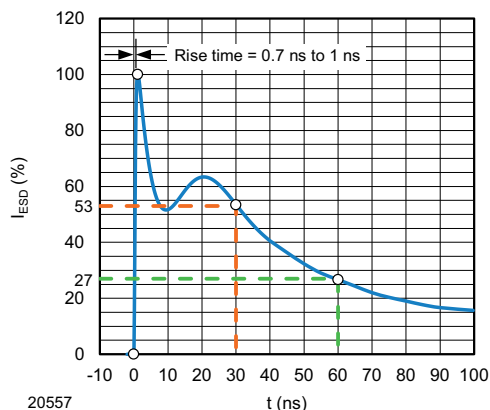


Fig. 1 - ESD Discharge Current Wave Form acc. IEC 61000-4-2 (330 Ω / 150 pF)

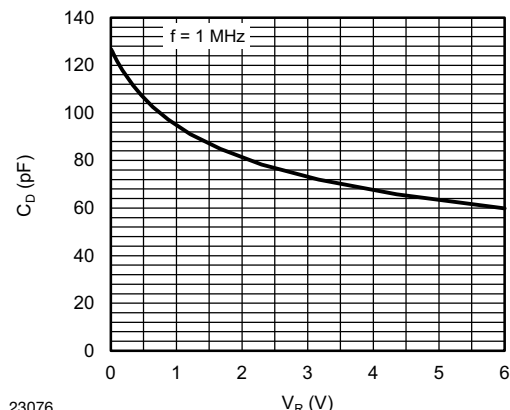


Fig. 4 - Typical Capacitance vs. Reverse Voltage

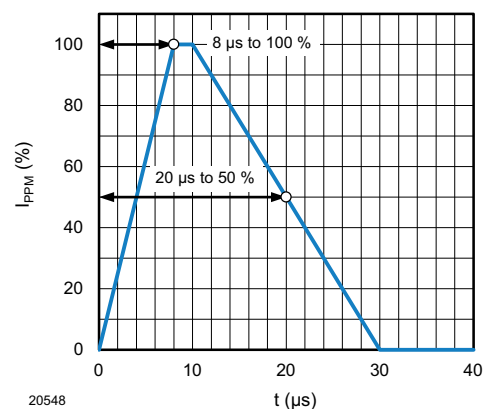


Fig. 2 - 8/20 μ s Peak Pulse Current Wave Form acc. IEC 61000-4-5

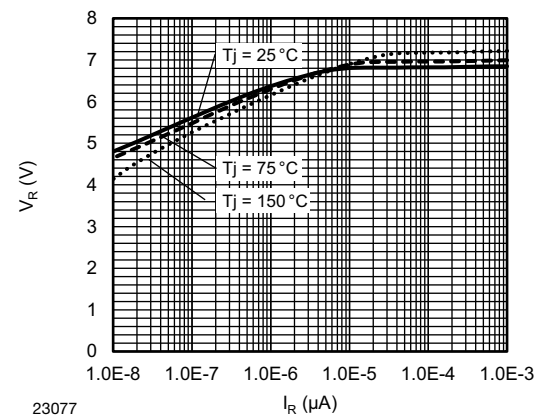


Fig. 5 - Typical Reverse Voltage vs. Reverse Current

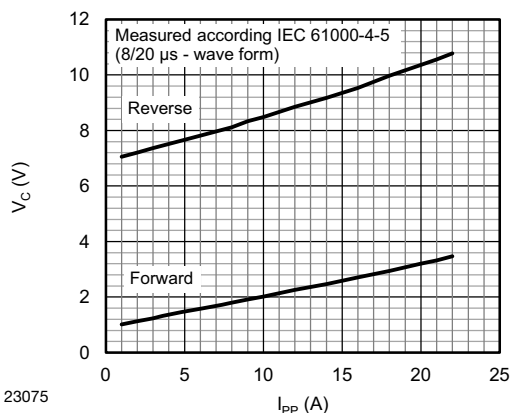


Fig. 3 - Typical Peak Clamping Voltage vs. Peak Pulse Current

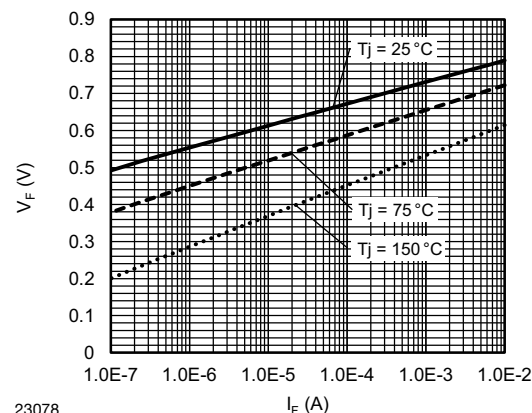


Fig. 6 - Typical Forward Voltage vs. Forward Current

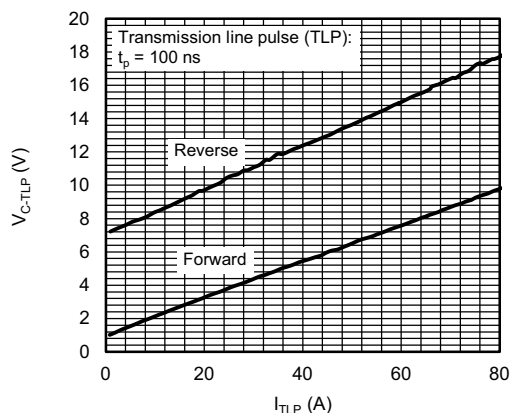
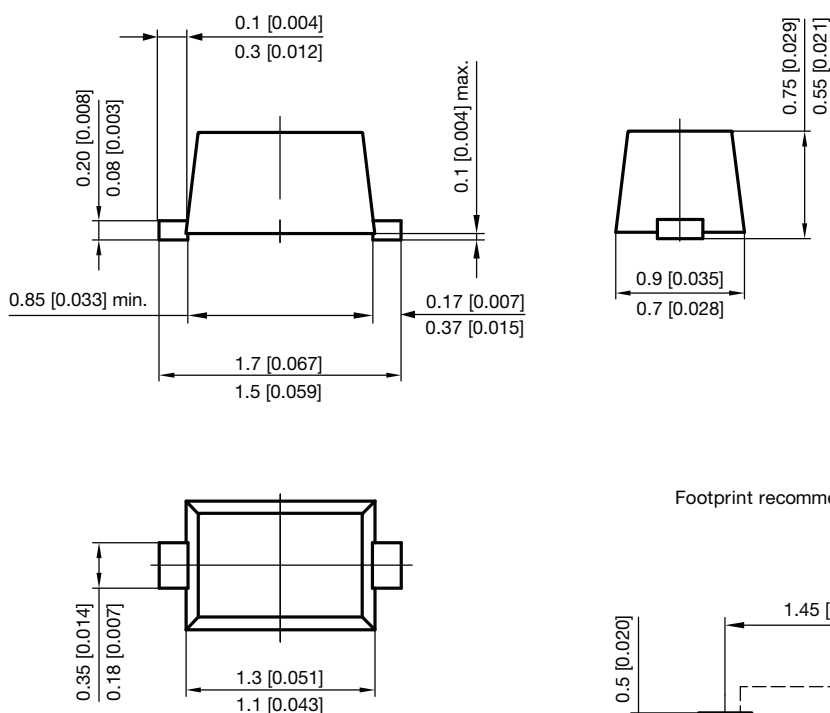


Fig. 7 - Typical Clamping Voltage vs. Peak Pulse Current

PACKAGE DIMENSIONS in millimeters [inches]: **SOD-523**


Footprint recommendation:

Document no.: S8-V-3880.02-003 (4)

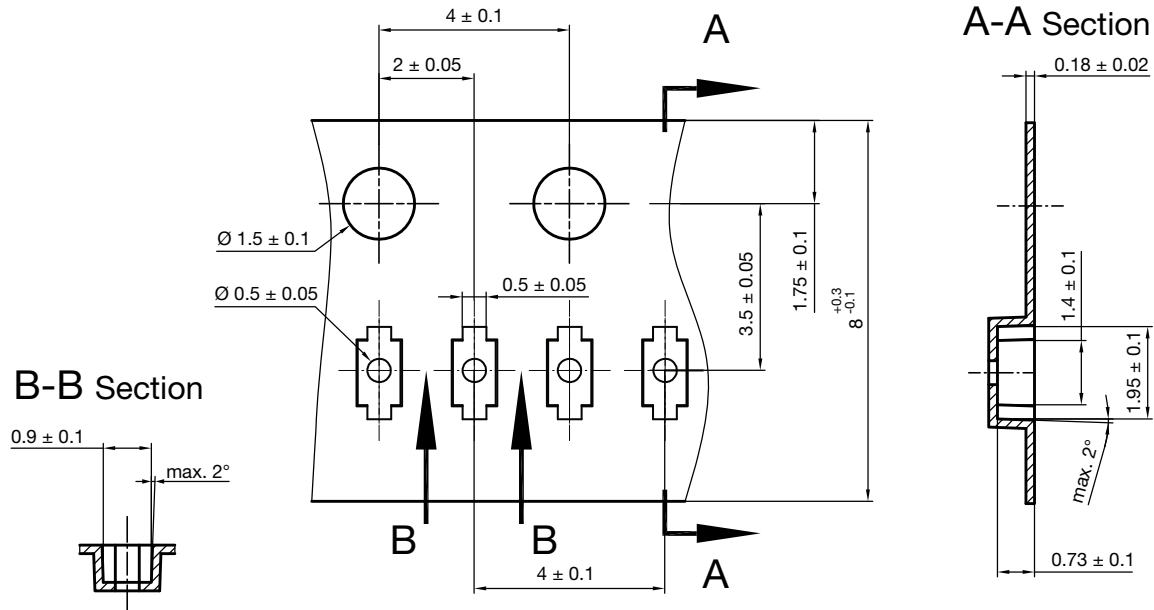
Created - Date: 04. April 2017

Rev. 4 - Date: 03. Aug. 2020

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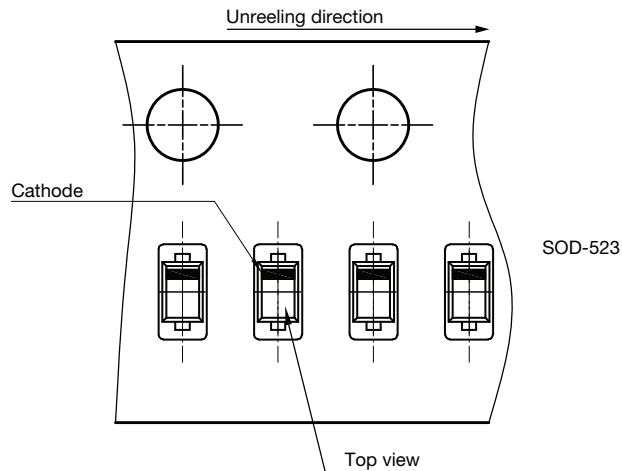


CARRIER TAPE SOD-523



S8-V-3717.03-005 (4)
05.07.2018
22959

ORIENTATION IN CARRIER TAPE SOD-523



S8-V-3717.03-006 (4)
05.07.2018
22958



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